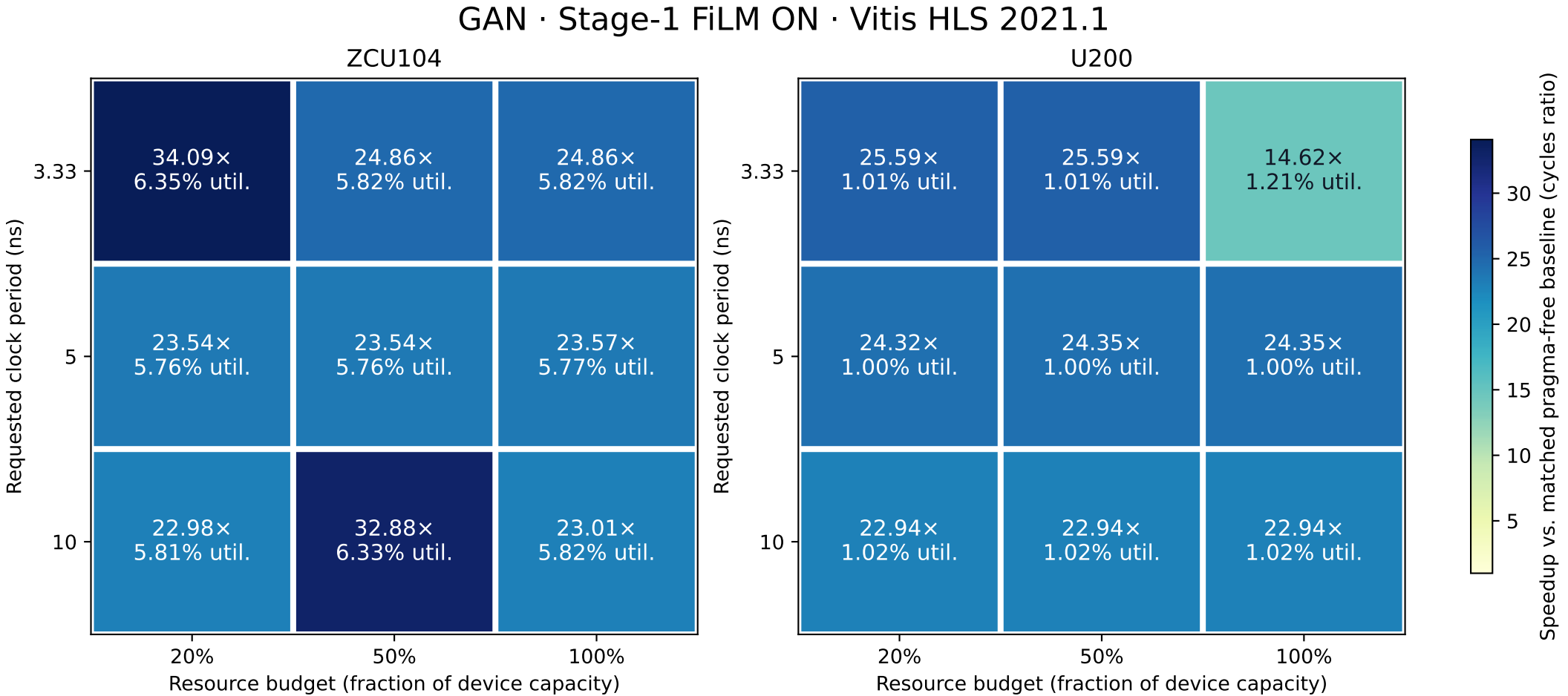

Mean utilization = average BRAM18K / DSP / FF / LUT usage relative to full device capacity.
† HLS estimated clock exceeds request. ‡ At least one resource exceeds the requested budget.
Nominal speedup uses matched requested periods; no post-route timing or functional simulation.